MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER PRACTICAL EXAMINATION

(2024 ADMISION ONWARDS)

MG1DSCIAM100 INTERACTIVE ROBOTIC SYSTEMS

Duration: 2 hours Maximum Marks: 35

Practical Examination – Model Question Paper

Instructions:

- **Demonstrate** the given task
- **Viva session** (Minimum 5 questions from the Practical Module)
- **Lab Report**

Evaluation Criteria:

- **Viva**: 10 marks
- **Lab Report**: 10 marks
- **Demonstration**: 15 marks

Tasks: (Choose any One/Two depending on the complexity)

1. Write a program to turn on and turn off LED
2. Write a program to create an SOS signal using LED
3. Control LED or a buzzer using LDR
4. Set up a light-controlled buzzer operation system
5. Design a parking Indicator using ultrasonic sensor
6. Create a smoke and fire alarm system
7. Assemble a robocar using geared DC motors and a Driver module
8. Write an Arduino program for PIR motion sensor-based burglar alarm using Wokwi simulator.
9. Write an Arduino program for PIR motion sensor-based burglar alarm and construct a circuit using Arduino Uno.
10. Write an Arduino program for LPG Gas leak detector using MQ2 sensor using Wokwi simulator.
11. Write an Arduino program for LPG Gas leak detector using MQ2 sensor and construct a circuit using Arduino Uno.
12. Gradually increase and decrease the intensity of LED use PWM function of Arduino.